Web Development Essentials Reference

Quick Reference Guide: Terminal Commands, Git Workflow, SemVer & VS Code Shortcuts

1. COMMAND LINE NAVIGATION & FILE MANAGEMENT

Command	Description & Usage
`pwd`	**Print Working Directory:** Displays full file path of your current folder.
`ls` / `ls -la`	**List Directory:** `ls` lists visible files. `-la` shows detailed list including hidden files.
`cd <dir>` / `cd ..`	**Change Directory:** Move into a specified folder, or use `..` to move up one folder.
`mkdir <folder>`	**Make Directory:** Creates a new folder. Use `-p` (e.g., `mkdir -p img/logo`) for nested folders.
`touch <file>`	**Create File:** Generates a new, empty file (e.g., `touch index.html css/style.css`).
`cp <src> <dest>`	**Copy:** Copies files. Use `-R` flag (`cp -R css csss`) to copy entire folders recursively.
`mv <src> <dest>`	**Move / Rename:** Renames a file or moves it to another folder (e.g., `mv about.html contact.html`).
`rm <file>`	**Remove File:** Permanently deletes a file. *Bypasses the system trash.*

2. GIT VERSION CONTROL WORKFLOW

Git Command	Action
`nano .gitignore` / `cat`	Edit ignored files list using terminal editor (`Ctrl+O` Save, `Ctrl+X` Exit). `cat` views contents.
`git init`	Initializes a new Git repository inside the current project folder.
`git status`	Checks tracking status of modified, staged, or untracked files.
`git add .`	Stages all new and modified files for the upcoming commit.
`git commit -m "msg"`	Saves staged changes permanently to history with a descriptive log message.
`git log`	Displays chronological log of past commits and author information.
`git branch <name>`	Creates a new parallel development branch (e.g., `git branch feature`).
`git checkout <target>`	Switches active branch or time-travels to a tag/commit (e.g., `git checkout main`).
`git merge <branch>`	Integrates changes from target branch into current active branch.
`git tag <version>`	Labels current commit state with a release marker (e.g., `git tag 0.1.0`).

3. SEMANTIC VERSIONING (SEMVER)

Standardized format for tracking software versions: `MAJOR.MINOR.PATCH` (e.g., `2.4.12`)

MAJOR (X.0.0)
Breaking Changes: Non-backward compatible API/code updates.

MINOR (0.X.0)
New Features: Added functionality in a backward-compatible manner.

PATCH (0.0.X)
Bug Fixes: Backward-compatible bug fixes and minor updates.

4. VS CODE PRODUCTIVITY & SHORTCUTS

Shortcut	Function
`Ctrl + P`	**Goto File:** Search and open files rapidly by name.
`Ctrl + W`	**Close File:** Closes the currently active editor tab.
`Ctrl + G`	**Goto Line:** Jump directly to a specific line number.
`Ctrl + D`	**Select Match:** Multi-selects next matching word for multi-cursor editing.
`Ctrl + Shift + P`	**Command Palette:** Access VS Code commands (e.g., *Join Line*).
`Alt + Up/Down`	**Move Line:** Moves current line or selection up or down.
`Shift + Alt + Down`	**Copy Line Down:** Duplicates current line downward.

5. EMMET SYNTAX & EXPANSION

> Pressing `Tab` after Emmet shortcuts expands them instantly: `!` creates a full HTML5 boilerplate, while `lorem` generates placeholder text.

Emmet Snippet Example

```
div#box>h3.title+ul>li*3>a[href=#]{Item $}
```

Expanded HTML Output

```
<div id="box">
  <h3 class="title"></h3>
  <ul>
    <li><a href="#">Item 1</a></li>
    <li><a href="#">Item 2</a></li>
    <li><a href="#">Item 3</a></li>
  </ul>
</div>
```